INTERNSHIP REPORT

Submitted by

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Enroll. No. :236170332009

In partial fulfillment for the award of the degree of

DIPLOMA ENGINEERING

in

Information and Communication technology

Govt. Polytechnic Ahmedabad College

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Industry Name: Gujarat Technological Technology

Internship Duration:22nd December 2025 to 30th March 2026

Academic Year:2025–2026

Government Polytechnic Ahmedabad
Nr. Panjarapole Ambawadi, Ahmedabad City – 380015

CERTIFICATE

This is to certify that the internship report submitted along with the project entitled **"Internship Report on Performance Analysis of NaVIC and GPS Using Different Satellite Parameters"** has been carried out by **Param Hingu** under my guidance in partial fulfilment for the Diploma Engineering in **Information and Communication Technology (ICT)** of **Gujarat Technological University, Ahmedabad** during the academic year **2025–26**.

Industry Mentor: **Head of Department:**

(Raj Hakani) **(Mr. Manish N. Charel)**

COMPANY CERTIFICATE

DECLARATION

I, **Param Jigneshkumar Hingu**, hereby declare that the internship report titled **"Performance Analysis of NaVIC and GPS Using Diffrent Satellite Parameters"** submitted to **Gujarat Technological University, Ahmedabad**, in partial fulfilment of the requirements for the Diploma in **Information and Communication Technology (ICT)** is a record of my own work.

This report is based on the internship carried out by me at **Atal Incubation Centre**, **Gujarat Technological University, Chandkheda, Ahmedabad** during the period from **22nd December 2025 to 30th March 2026**, under the guidance of my industry mentor.

I further declare that this report has not been submitted to any other university or institution for the award of any degree or diploma. I also confirm that all the information presented in this report is original and true to the best of my knowledge.

Place: Ahmedabad

Date:

Student Name: Param Jigneshkumar Hingu

Enrollment Number: 236170332009

Signature:

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Atal Incubation Centre, Ahmedabad**, for providing me with the opportunity to undergo my internship in the field of **Performance Analysis of NaVIC and GPS Using Diffrent Satellite Parameters**. This internship has been a valuable learning experience that helped me gain practical knowledge and industry exposure.

I am especially thankful to my **industry mentor, Mr. Raj Hakani**, for his continuous guidance, support, and encouragement throughout the internship. His practical insights and technical expertise helped me understand real-world application development and improved my problem-solving skills.

I would also like to extend my heartfelt thanks to the **Head of Department, Mr. Manish N. Charel**, and faculty members of **Government Polytechnic, Ahmedabad**, for giving me the opportunity to undertake this internship and for their constant support and motivation.

I am grateful to all the team members at Atal Incubation Centre. who directly or indirectly contributed to my learning and helped me during my internship period.

Finally, I would like to thank my family and friends for their encouragement and support, which motivated me to successfully complete this internship and prepare this report.

Param Jigneshkumar Hingu
Diploma in Information and Communication Technology (ICT)
Enrollment No: **236170332009**

ABSTRACT

This report presents a comprehensive overview of the work carried out during my internship at Atal Incubation Centre, Ahmedabad, in the domain of satellite navigation systems with a focus on comparative analysis of NaVIC and GPS. The internship was conducted over a period from 22nd December 2025 to 30th March 2026, with the objective of gaining practical understanding of global and regional navigation systems and their performance evaluation techniques.

During the internship, I studied various positioning parameters of GPS and NaVIC systems, particularly Dilution of Precision (DOP) metrics such as GDOP, PDOP, and HDOP. These parameters were analyzed to understand their impact on positioning accuracy and satellite geometry. I also explored how NaVIC performs in comparison to GPS under different conditions, highlighting its advantages in regional navigation and accuracy.

The primary objective of this internship was to bridge the gap between theoretical knowledge of satellite navigation and its practical analysis. Through this experience, I developed a deeper understanding of navigation system performance, data interpretation, and analytical skills related to positioning technologies.

The outcome of this internship was a detailed comparative study demonstrating the effectiveness of NaVIC over GPS based on DOP parameters, along with significant improvement in my technical knowledge and analytical capabilities. This experience has prepared me for future opportunities in the field of navigation systems and related technologies.

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CHAPTER I

INTRODUCTION

1.1 About the Organization

GTU Atal Incubation Centre (AIC) is an innovation and incubation hub established under Gujarat Technological University (GTU). The centre aims to promote research, innovation, and entrepreneurship among students by providing technical resources, mentorship, and a professional working environment.

AIC supports development in various technology domains such as software development, embedded systems, and communication technologies. It encourages students to work on real-world projects and enhances their practical knowledge beyond academic learning. The incubation centre provides an excellent platform for gaining hands-on experience and understanding industry practices.

1.2 Internship Domain / Field

The internship was carried out in the domain of real-time data analysis and satellite navigation systems using Python.

The primary focus was on studying and analyzing the performance of satellite-based navigation systems, specifically NavIC and GPS.

The work involved real-time data collection and evaluation of positioning accuracy using important parameters such as:

- **Position Dilution of Precision (PDOP)**
- **Horizontal Dilution of Precision (HDOP)**
- **Vertical Dilution of Precision (VDOP)**

Python programming was used for processing, analyzing, and visualizing the collected data.

The internship mainly focused on understanding navigation system performance and accuracy in real-world conditions.

1.3 Objectives of the Internship

The main objectives of this 6-month internship were:

- To understand the working of satellite navigation systems.
- To analyze and compare the performance of NavIC and GPS

-
- To evaluate positioning accuracy using PDOP, HDOP, and VDOP parameters.
- To develop skills in data processing, visualization, and interpretation.
- To improve problem-solving and analytical thinking abilities.
- To gain exposure to a professional and research-oriented work environment.
- To bridge the gap between academic learning and real-world applications.

1.4 Overview of the Project

During the internship, a project was carried out to compare the performance of NavIC and GPS based on real-time data analysis.

The project focused on evaluating positioning accuracy using DOP (Dilution of Precision) parameters such as PDOP, HDOP, and VDOP. These parameters help determine the reliability and precision of satellite navigation systems.

Based on the analysis, it was observed that NavIC provides better performance in certain regions, particularly within India, due to its regional satellite configuration. The project highlights the importance of satellite geometry in determining navigation accuracy and demonstrates the effectiveness of NavIC for regional applications.

CHAPTER 2

ORGANIZATION PROFILE

2.1 Overview of the Organization

GTU Atal Incubation Centre (AIC) is an incubation and innovation center established under Gujarat Technological University (GTU). It is supported by government initiatives to promote startups, research, and technological development among students and entrepreneurs.

The centre provides infrastructure, mentorship, and technical support to help individuals transform innovative ideas into real-world solutions. It focuses on domains such as software development, embedded systems, communication technologies, and data analysis.

AIC creates a professional and research-oriented environment where students can gain practical experience by working on live projects. During the internship, it provided an opportunity to work on real-time data analysis related to satellite navigation systems.

2.2 Activities / Work Areas

Instead of "services," AIC mainly focuses on the following activities:

1. **Research & Development (R&D)**
    - Work on innovative technical projects
    - Development of real-time systems
    - Experimental and analytical studies
2. **Startup Incubation**
    - Support for startup ideas
    - Mentorship and funding guidance
    - Business and technical assistance
3. **Technical Training**
    - Skill development programs
    - Workshops on latest technologies
    - Hands-on project-based learning
4. **Project Development**

    - Real-time industry-level projects
    - Data analysis and system evaluation
    - Collaboration with students and mentors
5. **Innovation Support**
    - Encouragement for new ideas
    - Prototype development
    - Problem-solving using technology

2.3 Organizational Structure

The organizational structure of AIC is designed to support innovation and efficient project execution:

- **Director / Head of AIC** – Overall management and strategic decisions
- **Project Mentors** – Guide students and monitor project progress
- **Technical Team** – Supports development and research activities
- **Startup Support Team** – Helps entrepreneurs and incubation projects
- **Interns / Students** – Work on assigned projects under guidance

During my internship, I worked under the guidance of my mentor and contributed to a project related to satellite navigation system analysis.

2.4 Technologies Used

During the internship, the following technologies and tools were used:

Programming Language

- Python

Tools & Platforms

- VS Code (Development Environment)
- Git & GitHub (Version Control)

Data Analysis

- NumPy
- Pandas
- Matplotlib (for visualization)

Domain-Specific Technologies

- Satellite navigation data processing
- DOP parameter analysis (PDOP, HDOP, VDOP)
- Real-time data evaluation

Navigation Systems Studied

- NavIC
- GPS

CHAPTER 3

WORK DONE / PROJECT DESCRIPTION

3.1 Problem Statement

Satellite navigation systems play a crucial role in modern positioning and tracking applications. However, the accuracy of these systems depends on various factors such as satellite geometry and signal quality.

Commonly used systems like **GPS** may show variations in accuracy depending on location and environmental conditions. On the other hand, India's regional navigation system **NavIC** is designed to provide better accuracy within the Indian region.

The problem is to evaluate and compare the performance of NavIC and GPS using real-time data and determine which system provides better positioning accuracy based on parameters like:

- Position Dilution of Precision (PDOP)
- Horizontal Dilution of Precision (HDOP)
- Vertical Dilution of Precision (VDOP)

3.2 Learning Phase and Understanding (Initial Difficulty)

During the initial phase of the internship, it was difficult to understand satellite navigation concepts and parameters such as PDOP, HDOP, and VDOP. These concepts required proper understanding of satellite geometry, signal behavior, and positioning accuracy principles.

It took approximately **2–3 weeks** to understand the working of navigation systems and interpretation of DOP parameters.

- **For learning and reference, the following resources were used:**

Educational Materials on GPS and NavIC Systems with Technical Documentation from NovAtel on GNSS Fundamentals

Reference:

- https://docs.novatel.com/OEM7/Content/Logs/NMEA_Standard_Logs_.htm

3.3 Data Processing & Data Handling Process

The overall data handling process was carried out in multiple stages:

Data processing was performed using Visual Studio Code with Python, where:

Distance Calculation Between Latitude and Longitude:

In this project, the distance between two geographical coordinates was calculated using the **Haversine Formula**, which is widely used in navigation systems such as GPS and NavIC.

Formula:

$$d = 2r \arcsin\left(\sqrt{\sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)}\right)$$

Where:

- ϕ\phiϕ = Latitude (in radians)
- λ\lambdaλ = Longitude (in radians)
- RRR = Earth's radius (6371 km)
- ddd = Distance between two points
- This formula enables accurate calculation of real-world distances between two points on the Earth's surface.

- PDOP, HDOP, and VDOP values were analyzed
- Comparative analysis between NavIC and GPS datasets was performed

5. Data Visualization

Graphs were generated using Python to compare:

- PDOP values
- HDOP values
- VDOP values

3.4 Methodology Used

The project was carried out using the following structured approach:

1. Data Collection

- Real-time navigation data was collected from satellite sources
- Data included PDOP, HDOP, and VDOP values

2. Data Preprocessing

- Cleaned and organized collected data
- Removed inconsistent or missing values

3. Data Analysis

- Compared NavIC and GPS performance
- Evaluated accuracy using DOP parameters
- Identified variations in positioning quality

4. Visualization

- Graphs and charts were created using Python libraries
- Compared PDOP, HDOP, and VDOP visually

5. Result Evaluation

- Analyzed which system provides better accuracy
- Observed real-time performance differences

3.5 Tools and Technologies Used

Programming Language

- Python

Libraries

- NumPy (numerical calculations)
- Pandas (data handling and analysis)
- Matplotlib (data visualization)

Development Tools

- Visual Studio Code
- Git & GitHub
- Chatgpt

Domain Concepts

- Satellite Navigation Systems
- DOP Parameters (PDOP, HDOP, VDOP)
- Real-time Data Analysis

3.6 Implementation Phase (Coding Work)

After understanding the concepts, approximately **2 weeks** were spent developing the Python-based implementation.

The coding phase included:

- Writing scripts for data processing
- Implementing Haversine formula for distance calculation
- Comparing NavIC and GPS datasets
- Generating graphs for visualization

This phase helped in converting theoretical knowledge into practical implementation.

3.7 System Design / Working

System Overview

The system is designed to analyze real-time navigation data and compare the performance of **NavIC** and **GPS**.

Working Steps

1. Input Data
- Real-time PDOP, HDOP, VDOP values collected
2. Processing
- Python scripts process data
- Distance calculations performed
3. Analysis
- Accuracy of both systems evaluated
- Performance differences identified
4. Output
- Graphical representation of results
- Final comparison and conclusion

System Modules

1. **Data Collection Module**

- Handles real-time input data

2. **Data Processing Module**

- Cleans and organizes data

3. **Analysis Module**

- Compares NavIC and GPS performance

4. **Visualization Module**

- Generates graphs for interpretation

CHAPTER 4

IMPLEMENTATION DETAILS

4.1 Modules Developed

The project was divided into several functional modules to ensure proper data handling, analysis, and result generation. The implementation mainly focuses on real-time data processing and comparison of satellite navigation systems.

1. Data Collection Module

The Data Collection Module is responsible for gathering real-time navigation data required for analysis. The data includes parameters such as Position Dilution of Precision (PDOP), Horizontal Dilution of Precision (HDOP), and Vertical Dilution of Precision (VDOP).

This data is obtained from satellite navigation sources for both **NavIC** and **GPS**. The module ensures that the collected data is relevant and suitable for further processing.

2. Data Preprocessing Module

The Data Preprocessing Module prepares the collected data for analysis. Real-time data may contain inconsistencies or missing values, so this module performs:

- Data cleaning
- Removal of invalid or null values
- Formatting and structuring of data

Python libraries such as Pandas are used to efficiently handle and organize the dataset before analysis.

3. Data Analysis Module

The Data Analysis Module is the core part of the project. It processes the cleaned data and performs comparison between NavIC and GPS.

The analysis is based on:

- PDOP (Position accuracy)
- HDOP (Horizontal accuracy)
- VDOP (Vertical accuracy)

Lower DOP values indicate better positioning accuracy. The module calculates and compares these values to evaluate system performance.

4. Visualization Module

The Visualization Module presents the analyzed data in graphical form for better understanding. Using Python libraries like Matplotlib, the system generates:

- Line graphs
- Comparative charts
- Parameter-based visual analysis

These visualizations help in clearly identifying which system performs better under different conditions.

4.2 Screenshots / Outputs

1. Data Input / Dataset View:

- **Static Raw Data(Multi-Gnss):**

```
$PQTMANTENNASTATUS,1,0,1*4F
$GNRMC,112745.076,V,,,,,,,070226,,,N,V*2D
$GNGGA,112745.076,,,,,0,0,,,M,,M,,*53
$GNVTG,,T,,M,,N,,K,N*32
$GNGSA,A,1,,,,,,,,,,,,,,,,,1*1D
$GNGSA,A,1,,,,,,,,,,,,,,,,,3*1F
$GNGSA,A,1,,,,,,,,,,,,,,,,,6*1A
$GPGSV,1,1,02,21,,,31,05,,,24,1*64
$GAGSV,1,1,00,7*73
$GIGSV,1,1,01,03,,,29,1*74
$GNGLL,,,,,112745.076,V,N*61
$PQTMANTENNASTATUS,1,0,1*4F
$GNRMC,112746.076,V,,,,,,,070226,,,N,V*2E
$GNGGA,112746.076,,,,,0,0,,,M,,M,,*50
$GNVTG,,T,,M,,N,,K,N*32
$GNGSA,A,1,,,,,,,,,,,,,,,,,1*1D
$GNGSA,A,1,,,,,,,,,,,,,,,,,3*1F
$GNGSA,A,1,,,,,,,,,,,,,,,,,6*1A
$GPGSV,1,1,02,21,,,32,05,,,25,1*66
$GAGSV,1,1,00,7*73
$GIGSV,1,1,01,03,,,30,1*7C
$GNGLL,,,,,112746.076,V,N*62
$PQTMANTENNASTATUS,1,0,1*4F
$GNRMC,112747.076,V,,,,,,,070226,,,N,V*2F
$GNGGA,112747.076,,,,,0,0,,,M,,M,,*51
$GNVTG,,T,,M,,N,,K,N*32
$GNGSA,A,1,,,,,,,,,,,,,,,,,1*1D
```

- **Static Raw Data(Neo-6m):**

```
$GPRMC,,V,,,,,,,,,,N*53
$GPVTG,,,,,,,,,,N*30
$GPGGA,,,,,,0,00,99.99,,,,,,*48
$GPGSA,A,1,,,,,,,,,,,,,99.99,99.99,99.99*30
$GPGSV,1,1,00*79
$GPGLL,,,,,,V,N*64
$GPRMC,,V,,,,,,,,,,N*53
$GPVTG,,,,,,,,,,N*30
$GPGGA,,,,,,0,00,99.99,,,,,,*48
$GPGSA,A,1,,,,,,,,,,,,,99.99,99.99,99.99*30
$GPGSV,1,1,00*79
$GPGLL,,,,,,V,N*64
$GPRMC,,V,,,,,,,,,,N*53
$GPVTG,,,,,,,,,,N*30
$GPGGA,,,,,,0,00,99.99,,,,,,*48
$GPGSA,A,1,,,,,,,,,,,,,99.99,99.99,99.99*30
$GPGSV,1,1,00*79
$GPGLL,,,,,,V,N*64
$GPRMC,,V,,,,,,,,,,N*53
$GPVTG,,,,,,,,,,N*30
$GPGGA,,,,,,0,00,99.99,,,,,,*48
$GPGSA,A,1,,,,,,,,,,,,,99.99,99.99,99.99*30
$GPGSV,1,1,00*79
$GPGLL,,,,,,V,N*64
$GPRMC,,V,,,,,,,,,,N*53
$GPVTG,,,,,,,,,,N*30
```

➢ Code Execution:

```python
from collections import defaultdict
import pandas as pd
import os
from openpyxl import load_workbook

# ------------------------------------------------
# Convert NMEA lat/long to decimal degrees
# ------------------------------------------------
def nmea_to_decimal(value, direction):
    if not value:
        return None
    try:
        if direction in ("N", "S"):
            degrees = int(value[:2])
            minutes = float(value[2:])
        else:
            degrees = int(value[:3])
            minutes = float(value[3:])

        decimal = degrees + minutes / 60

        if direction in ("S", "W"):
            decimal *= -1
```

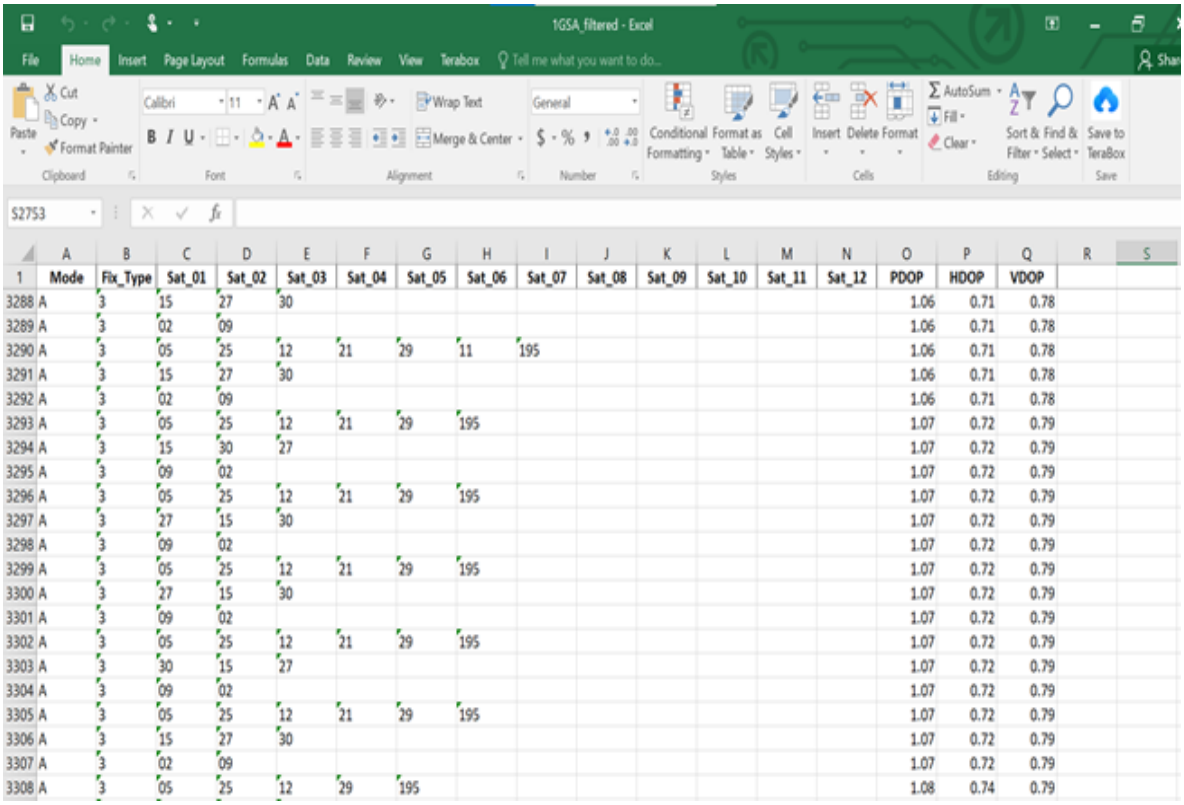
```python
def nmea_to_decimal(value, direction):

        return round(decimal, 6)
    except:
        return None


# ------------------------------------------------
# Group NMEA sentences
# ------------------------------------------------
def group_nmea(lines):
    grouped = defaultdict(list)

    for line in lines:
        line = line.strip()
        if not line.startswith("$"):
            continue

        sentence = line.split("*")[0]
        parts = sentence.split(",")
        sentence_type = parts[0][1:]
        grouped[sentence_type].append(parts)

```

```python
def group_nmea(lines):
    return grouped


# ------------------------------------------------
# Parsers
# ------------------------------------------------
def parse_gprmc(data):
    rows = []

    for d in data:
        # Ensure minimum required length
        while len(d) < 12:
            d.append("")

        rows.append({
            "Message ID": d[0],                         # GPRMC
            "UTC Time": d[1],                           # hhmmss
            "Status": d[2],                             # A / V
            "Latitude": nmea_to_decimal(d[3], d[4]),
            "Longitude": nmea_to_decimal(d[5], d[6]),
            "Speed (knots)": float(d[7]) if d[7] else None,
            "Course (deg)": float(d[8]) if d[8] else None,
            "Date": d[9],                               # ddmmyy
            "Magnetic Variation": float(d[10]) if d[10] else None,
            "Mag Var Direction": d[11]
        })

    return rows



def parse_gpgga(data):
    rows = []
    for d in data:
        rows.append({
            "Time (UTC)": d[1],
            "Latitude": nmea_to_decimal(d[2], d[3]),
            "Longitude": nmea_to_decimal(d[4], d[5]),
            "Fix Quality": d[6],
            "Satellites": d[7],
            "HDOP": d[8],
            "Altitude (m)": d[9]
        })
    return rows


def parse_gpvtg(data):
    rows = []

    for d in data:
        # Ensure minimum length
        while len(d) < 10:
            d.append("")

        rows.append({
            "Message ID": d[0],                         # GPVTG
            "True Track (deg)": float(d[1]) if d[1] else None,
            "True Track Ref": d[2],
            "Magnetic Track (deg)": float(d[3]) if d[3] else None,
            "Magnetic Track Ref": d[4],
            "Speed (knots)": float(d[5]) if d[5] else None,
            "Speed Unit (knots)": d[6],
            "Speed (km/h)": float(d[7]) if d[7] else None,
            "Speed Unit (km/h)": d[8],
            "Mode Indicator": d[9]
        })

    return rows
```

```python


def parse_gpgll(data):
    rows = []
    for d in data:
        while len(d) < 8:
            d.append("")
        rows.append({
            "message_id": d[0],
            "latitude_raw": d[1],
            "latitude_dir": d[2],
            "longitude_raw": d[3],
            "longitude_dir": d[4],
            "utc_time": d[5],
            "status": d[6],
            "mode_indicator": d[7],
            "latitude": nmea_to_decimal(d[1], d[2]),
            "longitude": nmea_to_decimal(d[3], d[4]),
        })
    return rows


# ------------------------------------------------
def append_df_to_excel(writer, book, df, sheet_name):
    if sheet_name in book.sheetnames:
        start_row = book[sheet_name].max_row
    else:
        start_row = 0

    df.to_excel(
        writer,
        sheet_name=sheet_name,
        index=False,
        header=(start_row == 0),
        startrow=start_row
    )


# ------------------------------------------------
# MAIN
# ------------------------------------------------
if __name__ == "__main__":

    input_file = "s.csv"                # Raw NMEA file
    output_file = "GPS_Output.xlsx"     # Excel output
```

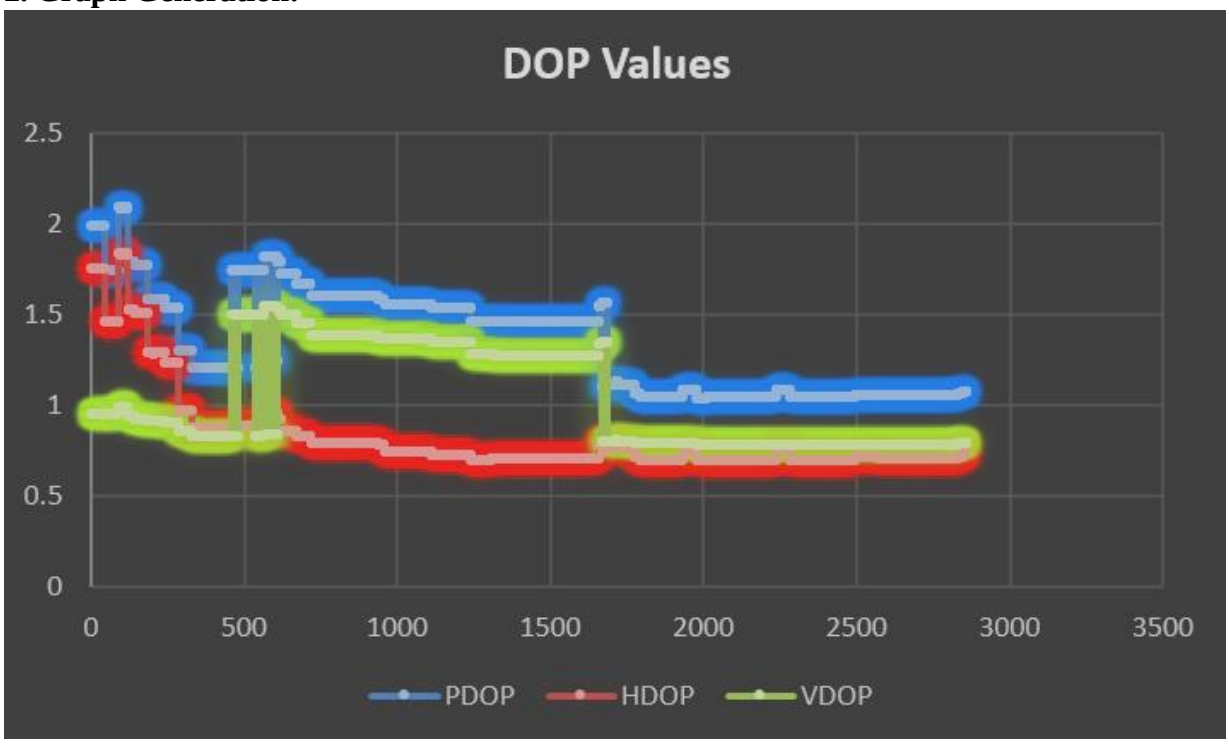
```python
    # Read raw file
    with open(input_file, "r") as f:
        lines = f.readlines()

    grouped = group_nmea(lines)

    # Create DataFrames
    df_gprmc = pd.DataFrame(parse_gprmc(grouped.get("GPRMC", [])))
    df_gpgga = pd.DataFrame(parse_gpgga(grouped.get("GPGGA", [])))
    df_gpvtg = pd.DataFrame(parse_gpvtg(grouped.get("GPVTG", [])))
    df_gpgll = pd.DataFrame(parse_gpgll(grouped.get("GPGLL", [])))
    df_gpgsa = pd.DataFrame(parse_gpgsa(grouped.get("GPGSA", [])))
    df_gpgsv = pd.DataFrame(parse_gpgsv(grouped.get("GPGSV", [])))

    # If Excel exists → append
    if os.path.exists(output_file):
        book = load_workbook(output_file)

        with pd.ExcelWriter(
            output_file,
            engine="openpyxl",
            mode="a",
            if_sheet_exists="overlay"
```

➢ **Dataset:**

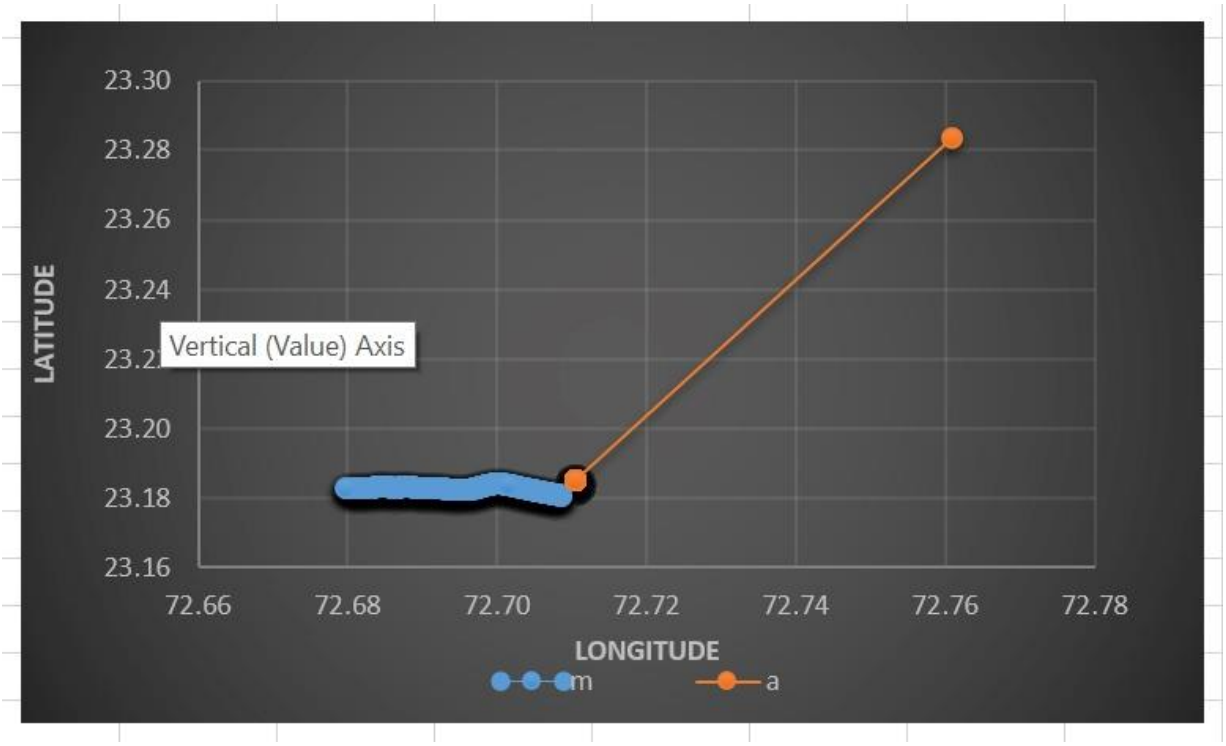


	Mode	Fix_Type	Sat_01	Sat_02	Sat_03	Sat_04	Sat_05	Sat_06	Sat_07	Sat_08	Sat_09	Sat_10	Sat_11	Sat_12	PDOP	HDOP	VDOP
3288	A	3	15	27	30										1.06	0.71	0.78
3289	A	3	02	09											1.06	0.71	0.78
3290	A	3	05	25	12	21	29	11	195						1.06	0.71	0.78
3291	A	3	15	27	30										1.06	0.71	0.78
3292	A	3	02	09											1.06	0.71	0.78
3293	A	3	05	25	12	21	29	195							1.07	0.72	0.79
3294	A	3	15	30	27										1.07	0.72	0.79
3295	A	3	09	02											1.07	0.72	0.79
3296	A	3	05	25	12	21	29	195							1.07	0.72	0.79
3297	A	3	27	15	30										1.07	0.72	0.79
3298	A	3	09	02											1.07	0.72	0.79
3299	A	3	05	25	12	21	29	195							1.07	0.72	0.79
3300	A	3	27	15	30										1.07	0.72	0.79
3301	A	3	09	02											1.07	0.72	0.79
3302	A	3	05	25	12	21	29	195							1.07	0.72	0.79
3303	A	3	30	15	27										1.07	0.72	0.79
3304	A	3	09	02											1.07	0.72	0.79
3305	A	3	05	25	12	21	29	195							1.07	0.72	0.79
3306	A	3	15	27	30										1.07	0.72	0.79
3307	A	3	02	09											1.07	0.72	0.79
3308	A	3	05	25	12	29	195								1.08	0.74	0.79

2. Graph Generation:

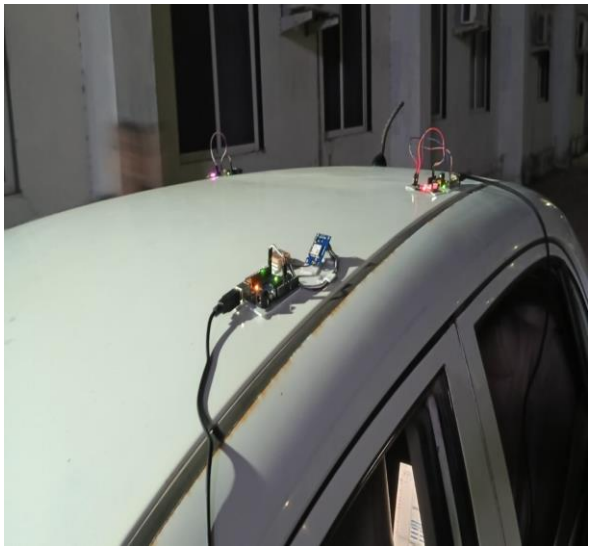


Output:

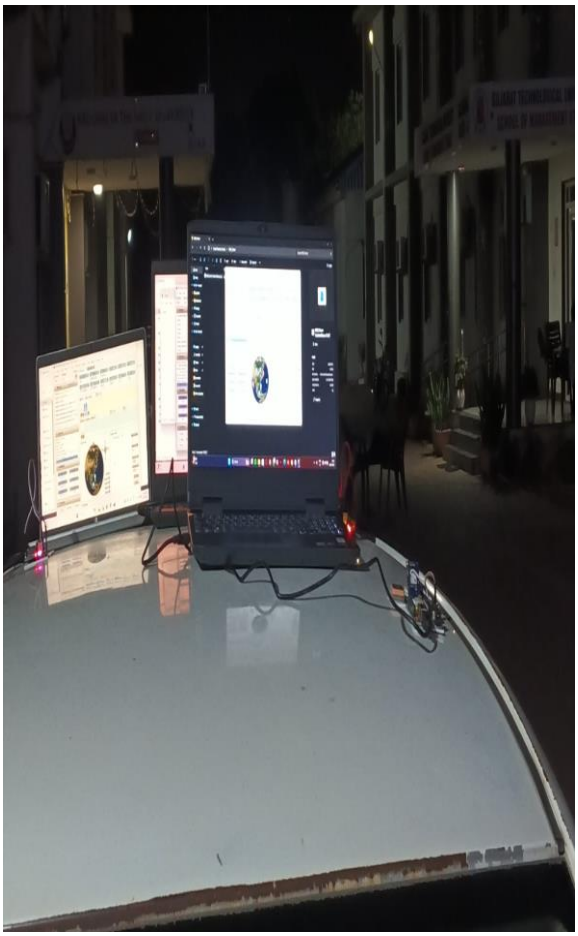

23.30
23.28
23.26
23.24
23.22
23.20
23.18
23.16
LATITUDE
Vertical (Value) Axis
72.66
72.68
72.70
72.72
72.74
72.76
72.78
LONGITUDE
m
a

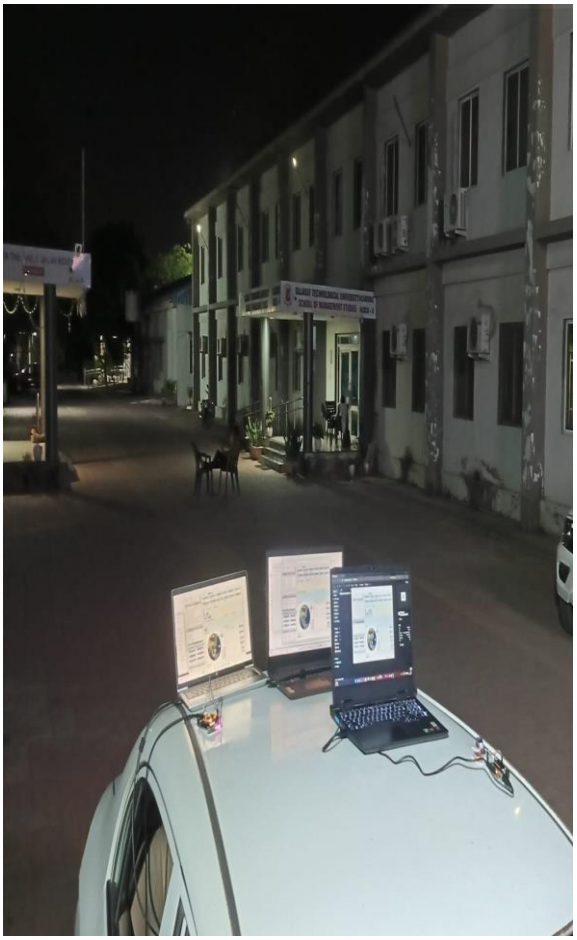
Dynamic Experiment:

Experimental Setup:



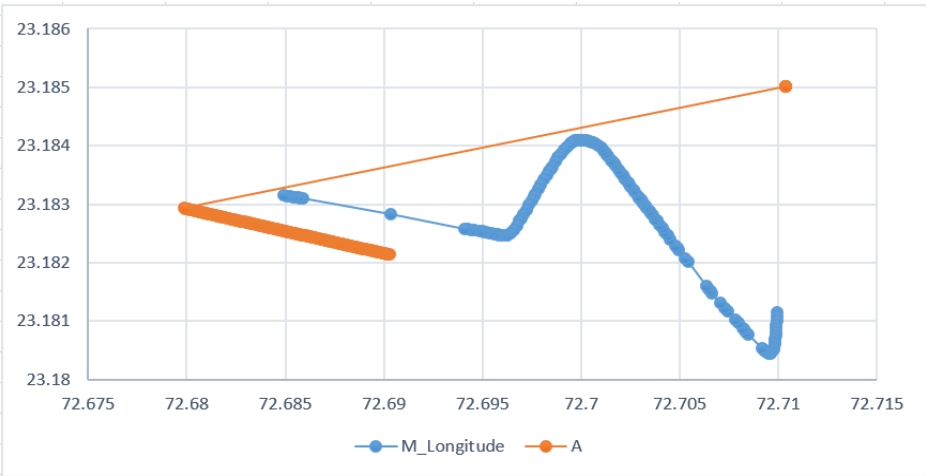
KA
KOI
JAWAB
NAHI
28/03/2026 22:14



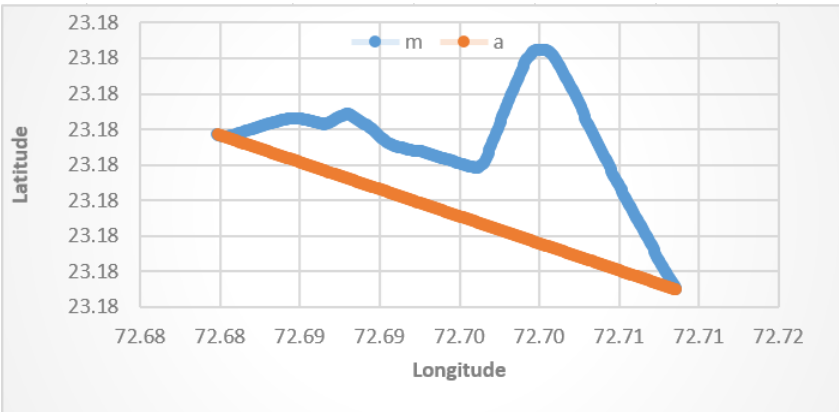


Grpah:

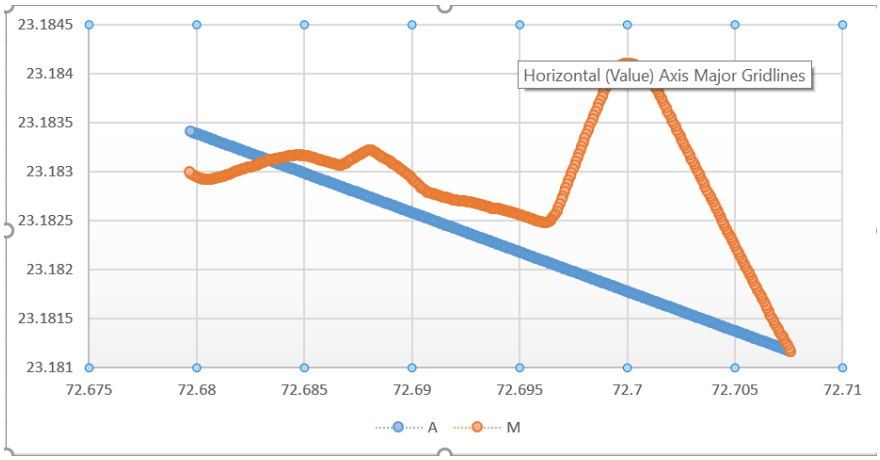
1. **MultiGNSS Receiver:**



2. **Gps Receiver:**



3. **NaViC Receiver:**



CHAPTER 5

RESULTS AND ANALYSIS

5.1 Outcomes Achieved

The developed system successfully performs real-time analysis of satellite navigation data for comparing **NavIC** and **GPS** based on DOP parameters.

The following outcomes were achieved during the internship project:

- Real-time navigation data was successfully collected and processed
- A structured dataset was created for PDOP, HDOP, and VDOP values
- Comparative analysis between NavIC and GPS was implemented
- Graphical visualization of navigation performance was generated using Python
- The system provides clear interpretation of positioning accuracy differences

The project demonstrates how real-time data analysis can be used to evaluate satellite navigation system performance effectively.

5.2 Performance / Results

The performance of the system was evaluated based on real-time data analysis and visualization results. The following observations were made:

- **NavIC generally shows lower DOP values compared to GPS**, indicating better accuracy in many cases
- PDOP, HDOP, and VDOP graphs clearly show variation between both systems
- The processing of data using Python was efficient and produced quick results
- Visualization using graphs helped in easy interpretation of performance differences
- The system successfully highlights the importance of satellite geometry in positioning accuracy

Overall, **NavIC performs better in regional conditions (India)** due to optimized satellite coverage compared to **GPS**.

5.3 Challenges Faced and Solutions

During the execution of the project, several challenges were encountered:

1. Data Collection Challenges

Problem:

Difficulty in obtaining consistent real-time navigation data for analysis

Solution:

Data was collected carefully and structured properly for PDOP, HDOP, and VDOP parameters.

2. Data Cleaning Issues

Problem:

Presence of missing or inconsistent values in dataset

Solution:

Handled using Python libraries like Pandas for cleaning and preprocessing data.

3. Data Visualization Complexity

Problem:

Difficulty in representing comparative results clearly

Solution:

Used Matplotlib to generate clear and well-labeled graphs for comparison.

4. Understanding DOP Parameters

Problem:

Initial difficulty in understanding PDOP, HDOP, and VDOP meaning

Solution:

Studied satellite navigation concepts and referred to technical documentation.

Refrence:

https://docs.novatel.com/

5. Performance Analysis Interpretation

Problem:

Interpreting which system performs better based on values

Solution:

Used rule: *Lower DOP value = Higher accuracy*, which simplified comparison.

CHAPTER 6

SKILLS LEARNED

6.1 Technical Skills

During the internship, I developed and enhanced several technical skills related to satellite navigation systems, data analysis, and software tools:

1. GNSS and Navigation Systems

• Gained understanding of Global Navigation Satellite Systems (GNSS) including GPS and NavIC

• Learned satellite positioning concepts such as PDOP, HDOP, and VDOP

• Understood satellite geometry, signal propagation, and positioning accuracy principles

• Performed comparative analysis between NavIC and GPS systems

2. Python Programming

• Improved understanding of Python syntax and logical problem-solving

• Applied Python for data processing and analysis related to navigation systems

• Developed scripts for handling and interpreting positional data

3. Data Analysis and Interpretation

• Analyzed satellite data and positioning parameters

• Interpreted accuracy metrics and performance differences between systems

• Evaluated navigation results for comparative study

4. Debugging and Problem Solving

• Identified and resolved technical issues during implementation

• Improved analytical thinking and troubleshooting skills

5. Version Control (Basic)

• Used Git and GitHub for managing project files

• Learned basic version control operations such as commit and update

CHAPTER 7

CONCLUSION

The internship at AIC (Atal Incubation Centre), GTU Chandkheda was a valuable learning experience that helped me bridge the gap between academic knowledge and practical understanding of satellite navigation systems. During this period, I gained in-depth knowledge of Global Navigation Satellite Systems (GNSS), particularly GPS and NavIC.

I successfully worked on the project involving comparative analysis of NavIC and GPS systems, where I understood key concepts such as satellite positioning, signal behavior, and parameters such as PDOP, HDOP, and VDOP. This experience helped me develop a strong conceptual foundation in navigation systems and improved my analytical and problem-solving abilities.

Throughout the internship, I also applied Python for data handling and analysis, which enhanced my programming skills in a practical context. The project provided exposure to real-world challenges in understanding positioning accuracy and system performance.

In addition to technical knowledge, I developed important professional skills such as communication, teamwork, time management, and adaptability, which are essential in a professional environment.

Overall, this internship significantly improved my technical knowledge, confidence, and understanding of real-world applications. It has motivated me to further explore the field of satellite navigation and emerging technologies.

CHAPTER 8

FUTURE SCOPE

The project on comparative analysis of NavIC and GPS systems provides a foundational understanding of satellite navigation and positioning accuracy. However, there are several areas where the work can be further improved and extended in the future:

8.1 Improvements Possible

• **Improved Data Accuracy Analysis:**

Incorporate more real-time and high-precision datasets to enhance the accuracy of analysis

• **Advanced Error Modeling:**

Study additional error sources such as atmospheric delays, multipath effects, and receiver noise for deeper analysis

• **Visualization Enhancements:**

Develop graphical representations such as plots and maps for better understanding of positioning errors and satellite geometry

• **Algorithm Optimization:**

Improve computational methods for faster and more efficient calculation of parameters like PDOP, HDOP, and VDOP

• **Integration of Additional GNSS Systems:**

Extend analysis to include other systems like GLONASS, Galileo, and BeiDou for a broader comparison

8.2 Extensions of the Project

• **Real-Time Navigation System:**

Develop a system that can process real-time satellite data and provide live positioning results

• **Mobile Application Development:**

Create a mobile application to demonstrate navigation accuracy and comparison between NavIC and GPS

• **AI-Based Accuracy Prediction:**

Use machine learning techniques to predict positioning errors and improve accuracy

• **Integration with IoT Devices:**

Apply the system in IoT-based applications such as vehicle tracking and smart transportation

• **Research and Development:**

Further extend the work for academic research or publication in the field of satellite navigation

Chapter 9

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